

TEST ON PROGRAMMING (LOOPS)

Solve any 6 (6x15 =90)

1. Write a program to take a program and check if it's Niven or perfect number. Based on user's choice.

A Niven number (or Harshad number) is an integer that is divisible by the sum of its digits.

A Perfect Number is a positive integer that is equal to the sum of its proper divisors (excluding itself).

2. Write a program to check if a number is spy number or special 2 digit number based on user's choice

A Spy Number is a number where the sum of its digits is equal to the product of its digits.

A Special 2-digit Number is a 2-digit number where the sum of its digits plus the product of its digits equals the original number.

3. Take a number and check if it's a prime palindrome number or not.

A Prime Palindrome is a number that is both a prime (divisible only by 1 and itself) and a palindrome (reads the same forward and backward).

4. Take marks of 30 students and calculate the class average, highest marks, lowest marks.

5. Take a number from user and print factorial of each digit. ($d! = 1 \times 2 \times 3 \times 4 \times \dots \times d$)

6. Take number from user and check if it's a duck number or pronic number based on user's choice.

A duck number is a positive integer that contains at least one zero, but does not start with zero.

A Pronic Number (or oblong number) is a number that is the product of two consecutive integers. Formally, a number n is pronic if it can be expressed as $n=k(k+1)$ for some integer k

7. Take a number and check if it's automorphic or tech number based on user's choice.

An Automorphic Number is a number whose square ends with the same digits as the number itself.

A Tech Number is a number with an even number of digits that can be split into two equal halves, and the square of the sum of these halves equals the original number.